

Blood Konnector Complete Website Functional Flow

Connecting Donors and Recipients for Life-Saving Impact

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Blood Konnector Team

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1 Website Entry & Registration Flow

1.1 Home Page

- Overview of the Blood Konnector mission, donorrecipient connection, and success statistics.
- Quick access buttons:
 - Become a Donor
 - Request Blood (Emergency / Lifeline)
 - Login / Register
 - About Us / Contact / FAQs

1.2 User Registration Options

1. Donor Registration

- *Required Fields:* Full Name, Age, Gender, Blood Group, City, Contact Number, Email, CNIC, Password.
- *Optional Fields:* Health Conditions, Donation Availability Schedule, Emergency Availability.
- *Verification:* Email / OTP Verification → Profile Activation.

2. Recipient Registration

- Option 1: Emergency Request
- Option 2: Join Lifeline Panel
- Fields depend on chosen type (explained below).

3. Login Page

- Donor / Recipient / Admin login with authentication and role-based dashboard access.

2 Emergency Panel Flow (Instant Blood Requests)

2.1 Purpose

Immediate donor-recipient matching in emergency cases such as accidents, surgeries, or transfusion needs within hours.

2.2 Step 1: Emergency Request Form

Fields:

- Full Name, Age, Gender
- Required Blood Group

- Hospital Name & Location (Geo Tag Optional)
- Doctor Name (optional)
- Units Required
- Contact Number
- Emergency Description
- Date / Time Needed

System Actions:

- Request logged into Emergency Requests Database.
- Auto-search triggers Donor Matching Algorithm (filter: blood group + city + availability = emergency yes).
- Matches displayed instantly to recipient.

2.3 Step 2: Donor Connection & Chat

- Recipient can initiate chat with matched donors.
- Once both parties confirm donation, the system records:
 - Confirmed Date, Time, Hospital.
 - Adds countdown timer to both dashboards.
- If donor or recipient doesn't respond within 12 hours, → Request Timeout notification sent.

2.4 Step 3: Donation Confirmation Flow

- Post-donation, the system asks recipient:
 - *Did you receive the blood?* (Yes / No / Reschedule)
- If Yes → adds to Successful Donation Record.
- If No → counts as Failed Donation.
- If Rescheduled → system restarts the confirmation loop with new date and time.

2.5 Step 4: Feedback Collection

After completion:

- *Recipient Feedback:* Ease of process, communication clarity, donor responsiveness.
- *Donor Feedback:* Recipient coordination, hospital management, overall experience.
- Both ratings go to the donor/recipient performance metrics in the admin panel.

3 Lifeline Panel Flow (Regular / Recurring Recipients)

3.1 Purpose

Connect patients with chronic conditions (Thalassemia, Cancer, etc.) to regular donors.

3.2 Step 1: Join Lifeline Panel (Form)

Recipient fills the Lifeline Join Form:

- Personal Info
- Medical Info
- Emergency & Verification Details
- Consent & Declaration

System Actions:

- Admin reviews and verifies documents (CNIC, Medical Letter).
- Once approved → Recipient profile added to Lifeline Panel database.

3.3 Step 2: Recipient Dashboard

Sections:

- Upcoming Donation Schedule (Countdown Timers)
- Preferred Donor List (Regular donors assigned)
- Request New Donation (button to initiate request)
- Donation History (success / failed / rescheduled logs)
- Feedback Section (submit donor review after each donation)

Notifications:

- Automatic Reminders 72 / 24 / 6 hours before scheduled donation.
- Missed Donation Alert after countdown expiry.

3.4 Step 3: Donor Matching Algorithm

System automatically matches recipients with loyal or verified donors based on:

- Blood Group
- Location radius (km)
- Availability schedule
- Past performance (success rate / response rate).

Admin can manually assign backup donors if no automatic match found.

3.5 Step 4: Confirmation Cycle

- Once recipient creates a new request:
 - System notifies donors → first to accept is assigned.
- Countdown timer starts for scheduled donation.
- After time passes:
 - System asks recipient for confirmation (Yes / No / Reschedule).
 - Updates donation record accordingly.

3.6 Step 5: Admin Monitoring (Back Office)

Admin can:

- View all Lifeline members.
- Approve / Reject join requests.
- See donation statistics (success rates, cancellations).
- Handle complaints / feedback.
- Suspend inactive donors / recipients.
- Export data for NGO / government reports.

4 Notifications & Timeout Handling

Event	Notification Target	Action / Timeout
Chat Closed	System → Recipient	Confirm donation status prompt
Recipient Confirms Donation	System → Donor	Confirm availability for given date/time/location
No Response (12 hrs)	System → Both	Marks as timed out → adds to incomplete records
Countdown Start	System → Both	Reminder 24 hrs & 6 hrs before
Donation Time Passed	System → Recipient	Did you get the donation? (Yes / No / Reschedule)
Reschedule Confirmed	System → Both	Starts new countdown
Feedback Request	System → Both	After completion request feedback

5 Feedback & Rating System

- Both parties submit feedback after each donation.
- 15 rating scale + text remarks.
- Admin can see average ratings to prioritize reliable donors for future matches.

6 Backend Architecture Overview

- *Database Tables:*
 - Users (roles: donor, recipient, admin)
 - Blood Requests (emergency, lifeline)
 - Confirmations / Countdowns
 - Notifications
 - Feedback
 - Logs (success / fail / rescheduled)
- *Core Modules:*
 - Matching Engine
 - Chat Service (secured socket or Firebase)
 - Notification Scheduler (CRON / Job System)
 - Admin Analytics & Dashboard

7 Analytics & Reporting

Admin Dashboard Widgets:

- Total Donations (success, failed, rescheduled)
- Active Donors / Recipients
- Emergency vs Lifeline ratio
- Top performing donors
- City-wise blood demand map

8 Security & Verification

- CNIC + Email Verification for all users.
- Secure data storage (PII encrypted).
- Admin-only access for medical documents.
- Fraud detection on fake requests / spam accounts.